

Presentation review - rendered by PowerPoint at 1600 x 1200

slide 1 [KEEP] title



Make My Figure

One dataset, many controlled views - publication figures with the researcher in control

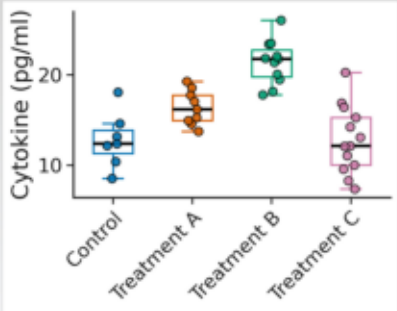
Desktop tutorial and demonstration
Version 1.1.1 - 39 plot types
Suresh Poudel

slide 2 [REDESIGN] motivate the problem



A real table rarely arrives ready to plot

col_A	measurement_2	condition_code	thing	score_final	id_value
item_01	2.644	C0	label1	12.45	1001
item_02	0.998	C0	label2	12.74	1002
item_03	2.883	C0	label3	12.58	1003
item_04	2.491	C0	label4	13.12	1004
item_05	2.49	C0	label5	16.47	1005
item_06	3.118	C0	label6	15.26	1006



Column names like col_A and measurement_2 are fine: you tell the application what each column means, and the figure follows.

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slide 3 [SPLIT] what the application is



What Make My Figure is

- A desktop application (and a browser version) that draws 39 publication-style plot types from CSV, TSV and Excel tables
- Runs entirely on your computer - no upload, no account
- Proposes column roles, figures and tests; every proposal can be changed
- Statistics on the figure: 19 tests, corrections, brackets, effect sizes, a methods sentence
- Individual observations with controllable size, jitter, fill, edge and arrangement
- Publication presets previewed on your own data before they are applied (experimental library)
- Reproducibility: PlotSpec, StatsSpec, Figure Package (.mmfpackage) with frozen data and checksums
- Multi-panel Figure Builder; open source, Python, Matplotlib and Qt

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slide 4 [KEEP] design principle



The application suggests. The researcher decides.

It proposes

Detects column types, recommends figures with a match score, proposes a column for every role, suggests a test from the design.

You decide

Every role is a drop-down of your own columns. Change any of them; the figure follows. Choose the test; the panel runs and documents it.

It records

PlotSpec for the specification, StatsSpec for the test, Figure Package for specification plus frozen data. Nothing recomputed behind your back.

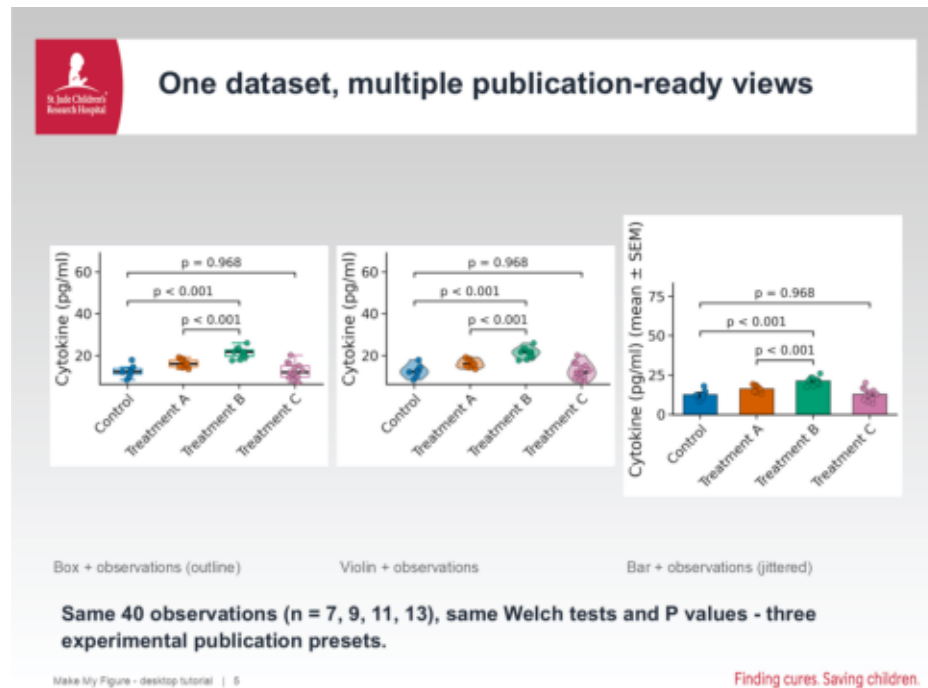
"The data do not belong to one plot."

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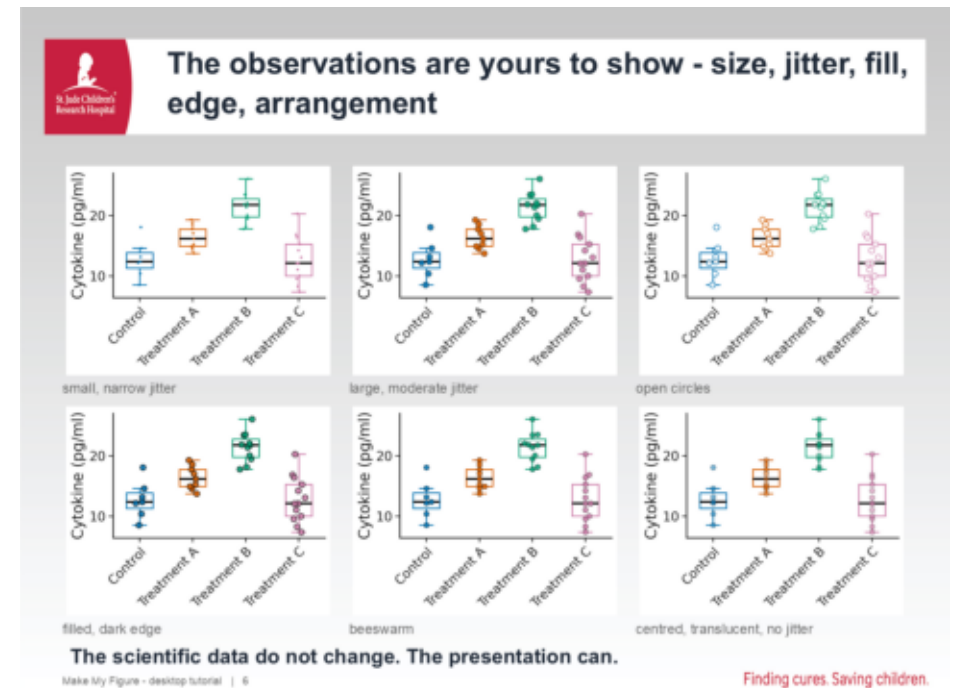
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slide 5 [REDESIGN] demo 1 - table with unhelpful names

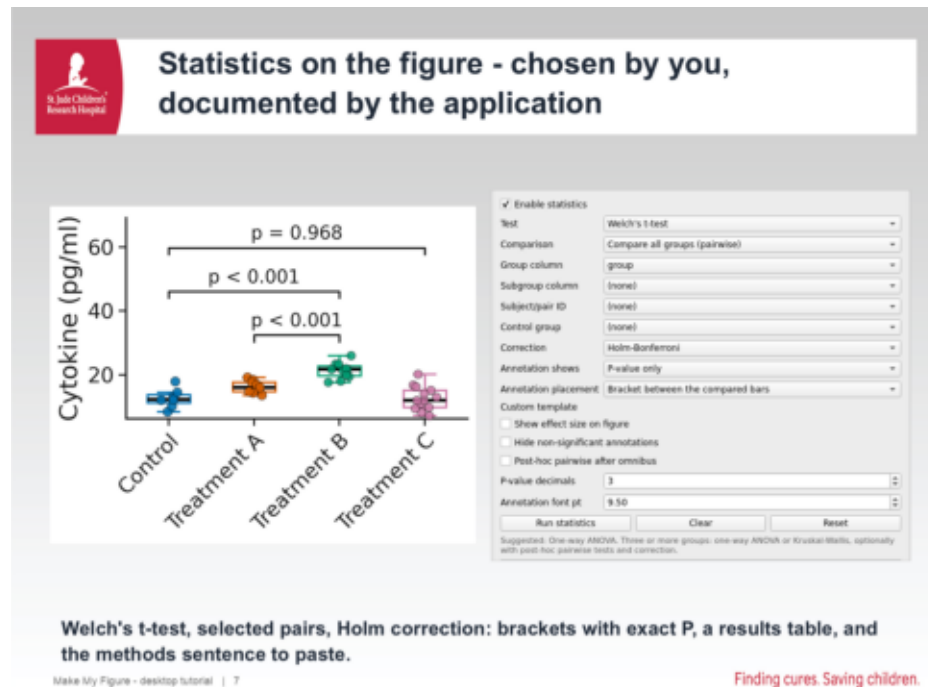


slide 6 [REDESIGN] demo 1 - correction and second plot

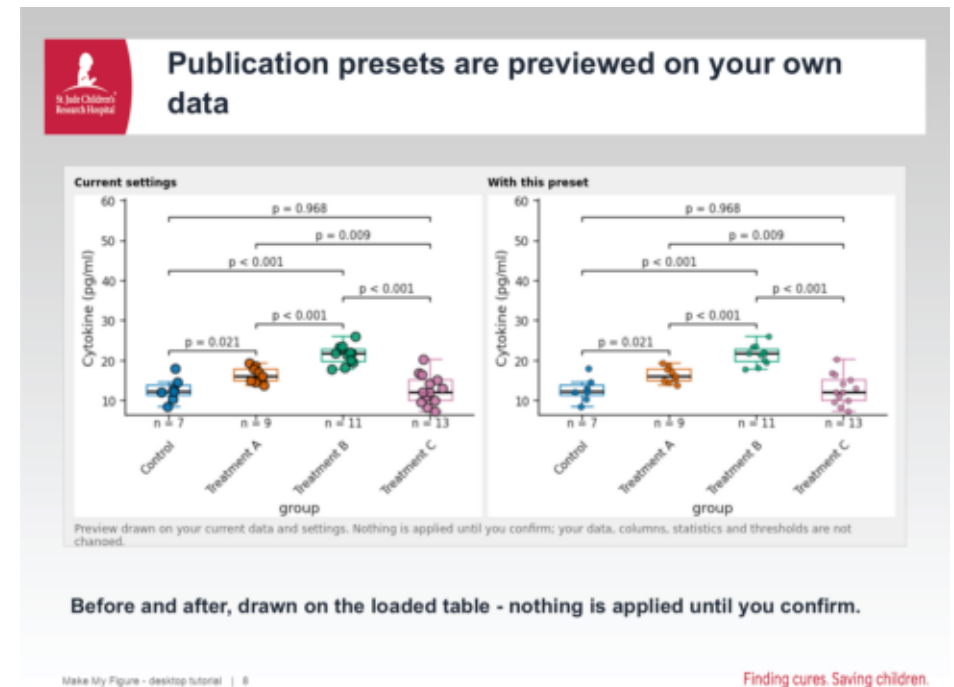


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slide 7 [SPLIT] demo 2 - volcano manual mapping



slide 8 [MINOR FIX] statistics on the figure



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slide 9 [REDESIGN] figure presets



The preview lists what changes and checks what must not

style.marker_alpha: None -> 0.9
mapping.point_edge_width: 1.0 -> 0.4
mapping.point_size: 60.0 -> 0.0

Experimental presets describe visual conventions measured in published open-access figures and the publishers' stated artwork requirements. They are not official journal templates, carry no endorsement, and do not guarantee acceptance. They change only how a figure is drawn, never your data, statistics or thresholds. Preview before applying.

Evidence family: group-comparison corpus (all three families)
Summary: biological_group_comparison_corpus.csv: bar/box/violin/dot panels coded for representation, markers, error bars, statistics and colour

Checked: no data, column role, statistical test, threshold or transformation changes.

Save as my preset...

Cancel

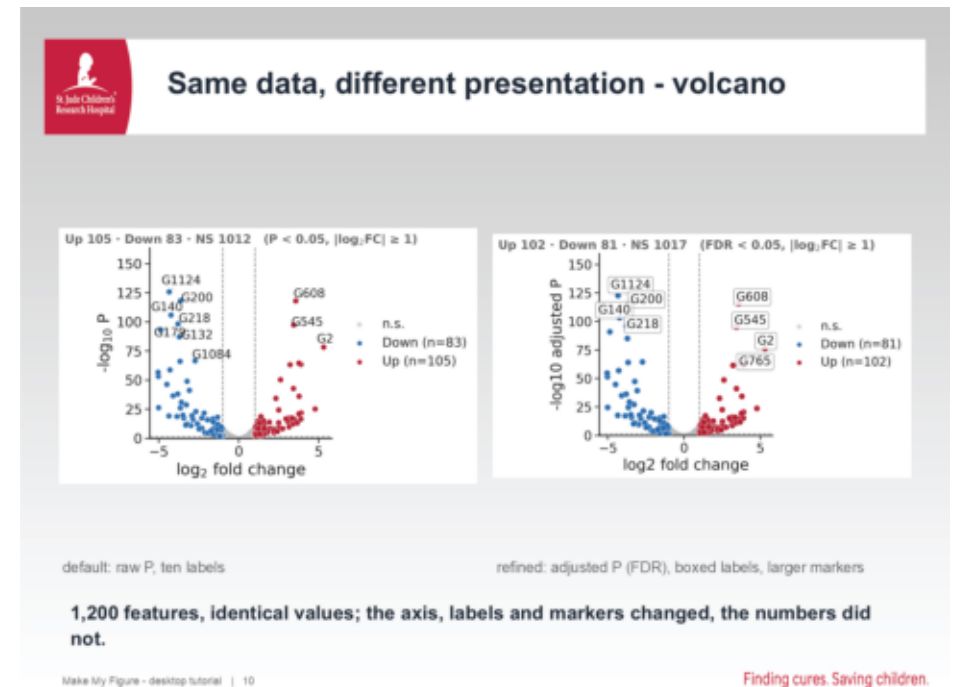
Apply

- Twelve experimental presets derived from published open-access figures and publishers' stated requirements; the letter in a name, (N) (S) (C), is the evidence set, not an approval
- Every setting that would change is listed (here marker opacity, point edge width, point size)
- Green line: no data, column role, statistical test, threshold or transformation changes - a preset that would be refused
- Apply, Cancel, or Save as my preset - a preset is a starting configuration, not a locked template

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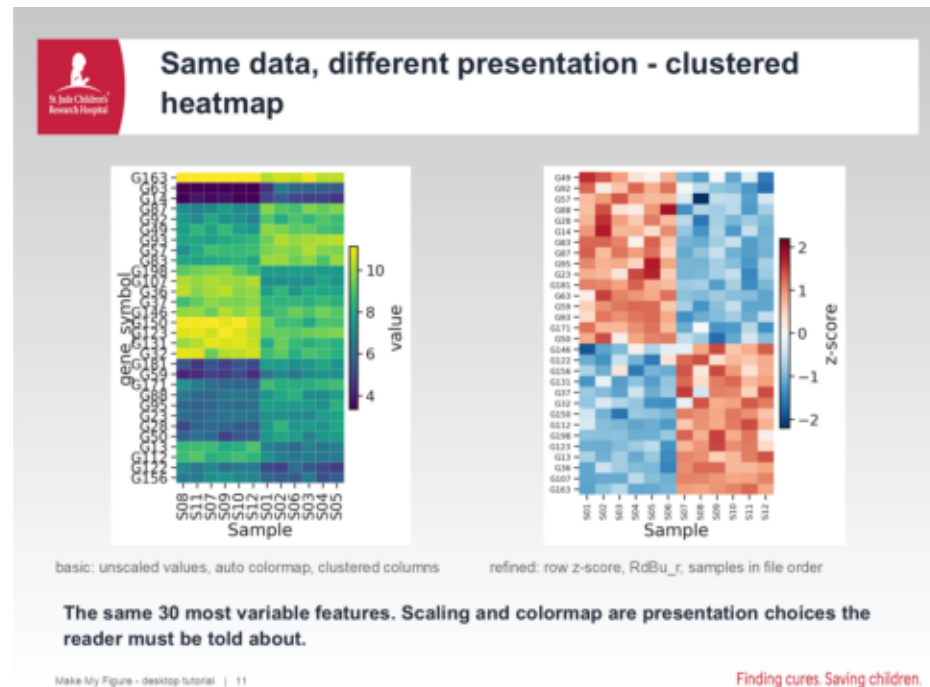
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slide 10 [REDESIGN] reproducibility

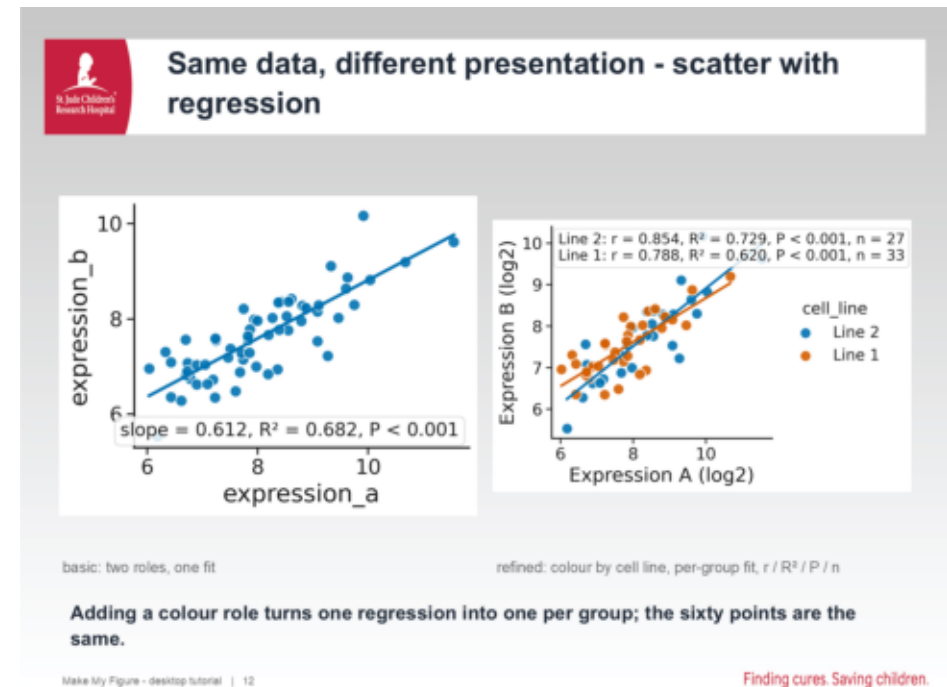


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slide 11 [MINOR FIX] figure builder

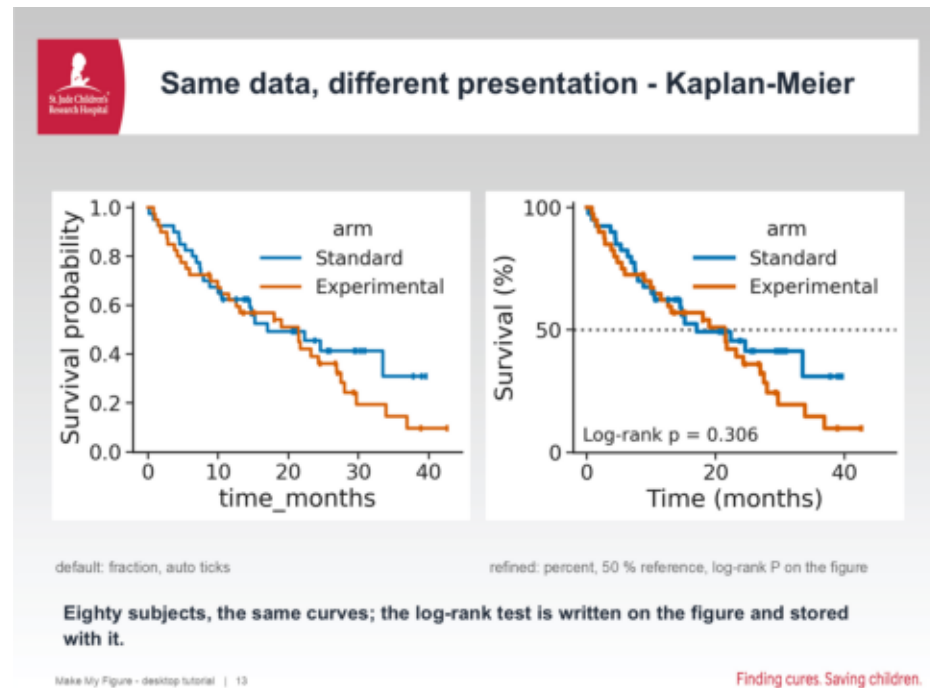


slide 12 [SPLIT] tutorial system

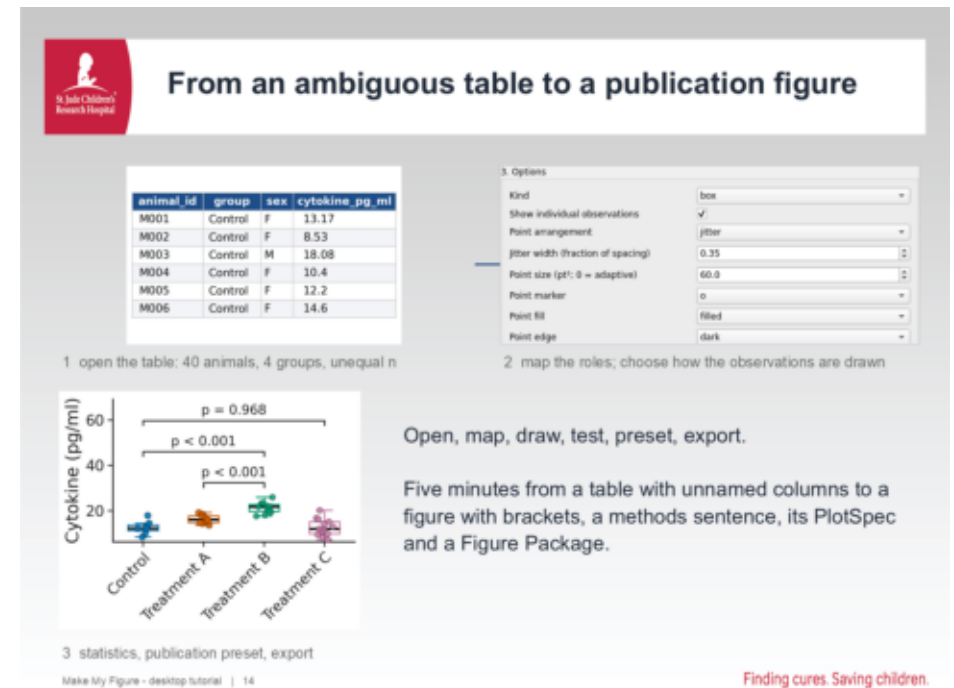


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slide 13 [KEEP] pilot validation table



slide 14 [KEEP] 39 plot types mosaic



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slide 15 [KEEP] roadmap



When the detector does not know your headers, you assign the roles

⚠ Your uploaded data does not have the columns required for "Volcano plot".
volcano_plot: missing required column(s) ["log2_fold_change", "p_value"]. Available columns: ["feature", "name", "mean_expr", "effect_measure", "t152", "p_raw", "p_adjusted", "cell", "annotation"]

Load Volcano plot example data

2. Map columns

x (none)

p (none)

label (none)

id_col (none)

every role starts at (none): the detector does not know these headers

2. Map columns

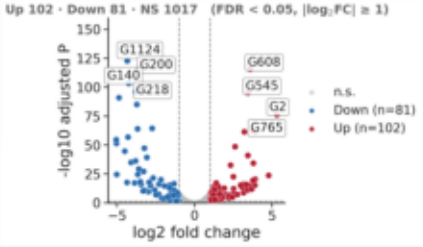
x effect_measure

p p_raw

label name

id_col feature

x, p, label, id_col assigned by hand



Up 102 · Down 81 · NS 1017 (FDR < 0.05, |log₂FC| ≥ 1)

-log₁₀ adjusted P

log₂ fold change

n.s.
Down (n=81)
Up (n=102)

The Messages tab names the missing roles; four drop-downs later the volcano renders - P values read from the table, never recomputed.

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slide 16 [KEEP] summary



Reproducibility - a Figure Package carries the figure and its frozen data

Figure packages include the data required to reproduce the figure.
This figure package contains the data required to reproduce the figure: a frozen copy of every table the plot was drawn from (and, when available, the original source file), the plot specification, statistics, preprocessing records and preview images. Share it only with people who may see these data.

This package will contain

- source table: group_comparison.csv (30 x 5) + original file
- records: PlotSpec, StatsSpec (results included)
- previews: PNG, SVG, PDF

Estimated size before compression: about 2.0 MB.

Save the package?

Save Cancel

• What is inside one .mmfpackage

- plot_spec.json, stats_spec.json (settings and results), render_metadata.json
- data/source.csv and the typed table; the original file when available
- preview/figure.png, .svg, .pdf; environment/environment.json
- manifest.json with a SHA-256 for every member - verified before drawing

• PlotSpec alone is the recipe; it needs the data file to redraw

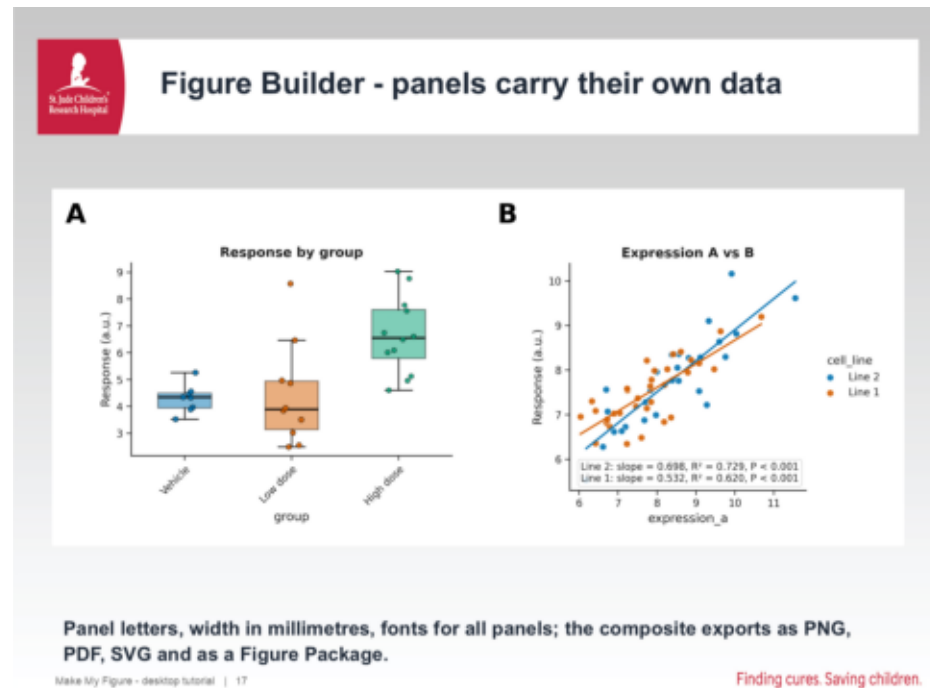
Move the one file anywhere, open it: integrity verified, mapping, options and brackets restored.

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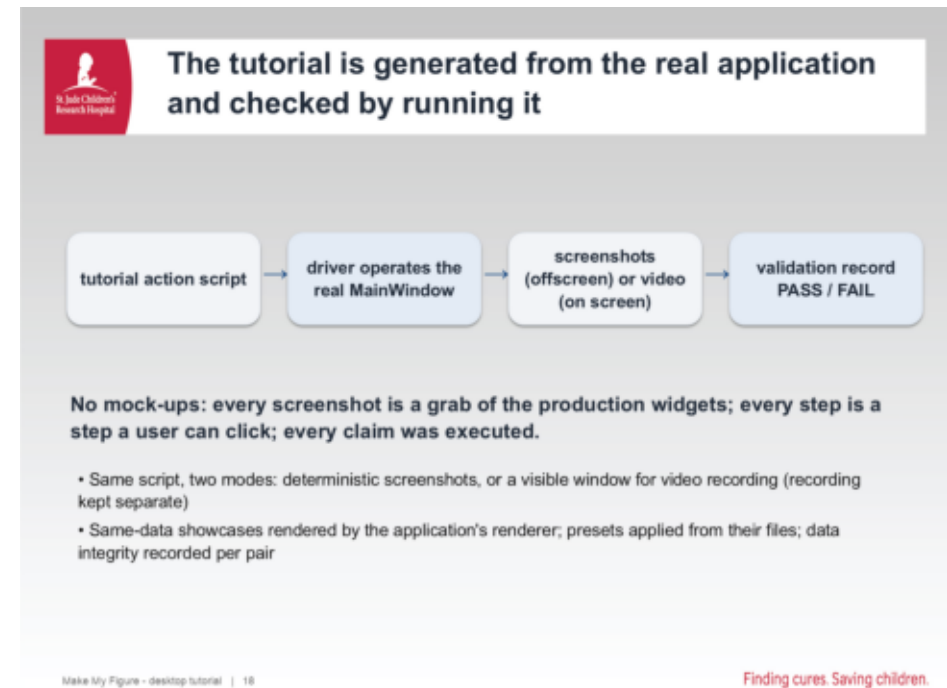
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Every tutorial instruction executed against the application

Tutorial	Status	Checks
Figure Builder: a two-panel composite	PASS	8 checks, 3 captures
Reproducibility: PlotSpec and Figure Package	PASS	9 checks, 7 captures
Figure presets: style travels, data stays	PASS	8 checks, 6 captures
Getting started with the desktop application	PASS	0 checks, 6 captures
Box / violin plot with points and statistics	PASS	8 checks, 8 captures
Clustered heatmap from a feature-by-sample matrix	PASS	5 checks, 6 captures
Kaplan-Meier survival curve	PASS	6 checks, 7 captures
Column mapping with ambiguous column names	PASS	7 checks, 11 captures
Showing individual observations and controlling jitter	PASS	20 checks, 17 captures
Publication presets: preview, apply, keep adjusting	PASS	13 checks, 9 captures
Scatter plot with regression line	PASS	6 checks, 8 captures
Volcano plot: automatic detection and manual role mapping	PASS	8 checks, 10 captures

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39 plot types, one workflow



Thumbnails for breadth only - each is a real render of the bundled synthetic example (tutorialPLOT_GALLERY.png)

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Where this goes next

- **Reviewed and rebuilt**
 - tutorial and slides re-evaluated as visual demonstrations; observations, jitter and publication presets shown at readable size
- **After author review**
 - remaining plot tutorials with the same template; matrix workflow tutorial and video; master and plot videos recorded live
 - decision whether the presets branch (observation controls, experimental presets) is merged - without it these chapters cannot ship
- **Not merged, tagged, released, published or pushed**

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Summary

- Open any table; the application detects, recommends and proposes - and you assign every role yourself
- One dataset, many controlled representations: summary plus every observation, with the test on the figure
- Publication presets are previewed on your data and remain a starting point
- PlotSpec carries the recipe; a Figure Package carries recipe and ingredients, verified
- The tutorial is generated from the real application and validated by running it

github.com/surPoudel/make-my-figure - tutorial/ on the pilot branch (not yet published)

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